PART A — QUESTIONS

AF19-001
Question:
Solve: $21x - 15 = 111$
A) 5
B) 6
C) 7
D) 8

AF19-002
Question:
If $19n + 8 = 141$, what is n?
A) 5
B) 6
C) 7
D) 8

AF19-003
Question:
Simplify: $6(5x - 3) - 8x$
A) $22x - 18$
B) $30x - 18$
C) $22x - 3$
D) $38x - 18$

AF19-004
Question:
A company charges $38 per job plus a $30 setup fee. Which expression represents the total charge for j jobs?
A) $38j + 30$
B) $30j + 38$
C) $68j$
D) $38 - 30j$

AF19-005
Question:
If $f(x) = 16x - 41$, what is $f(8)$?
A) 77
B) 87
C) 128
D) 169

AF19-006
Question:
Which ordered pair satisfies the equation $y = 14x + 2$?
A) (1, 15)
B) (2, 30)
C) (3, 44)
D) (4, 56)

AF19-007
Question:
Solve: 22(x - 5) = 154
A) 10
B) 11
C) 12
D) 13

AF19-008
Question:
A pattern begins: 8, 26, 44, 62, ... What is the next number?
A) 78
B) 79
C) 80
D) 81

AF19-009
Question:
Solve: x/35 = 3
A) 38
B) 70
C) 105
D) 140

AF19-010
Question:
A line has a slope of -16 and crosses the y-axis at 13. Which equation represents the line?
A) y = 13x - 16
B) y = -16x + 13
C) y = 16x + 13
D) y = -13x + 16

AF19-011
Question:
If 8 bins contain 264 parts, how many parts are in 5 bins at the same rate?
A) 150
B) 160
C) 165
D) 170

AF19-012
Question:
Solve: 240 - 26x = 136
A) 3
B) 4
C) 5
D) 6

AF19-013
Question:
Which expression is equivalent to 90x - 43x + 12?

A) 47x + 12
B) 133x + 12
C) 47x - 12
D) 90x - 31

AF19-014
Question:
A storage shelf starts with 1,300 parts. Each order removes 65 parts. Which equation gives the number of parts P left after o orders?
A) P = 1,300o - 65
B) P = 65o + 1,300
C) P = 1,300 - 65o
D) P = 65o - 1,300

AF19-015
Question:
If y = -20x + 260, what is y when x = 11?
A) 40
B) 50
C) 220
D) 480

AF19-016
Question:
Solve: 32x - 50 = 21x + 60
A) 8
B) 9
C) 10
D) 11

AF19-017
Question:
Which value of x makes 17x - 6 greater than 164?
A) 9
B) 10
C) 11
D) 8

AF19-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: -9, 10, 29, 48

What is the rule?
A) y = 19x - 9
B) y = 9x + 19
C) y = -9x + 19
D) y = 19x + 9

AF19-019
Question:
Solve: $16(x + 5) = 13x + 116$
A) 10
B) 11
C) 12
D) 13

AF19-020
Question:
If $a = -13$ and $b = 11$, what is $3a + 4b$?
A) -83
B) -5
C) 5
D) 83

AF19-021
Question:
Which point is located in Quadrant III?
A) (15, -8)
B) (-15, 8)
C) (-15, -8)
D) (15, 8)

AF19-022
Question:
A worker packs 210 items in 14 minutes. At the same rate, how many items can the worker pack in 9 minutes?
A) 120
B) 125
C) 135
D) 150

AF19-023
Question:
Solve for r: $B = 30r + 90$
A) $r = B - 90$
B) $r = (B - 90)/30$
C) $r = 30(B - 90)$
D) $r = B/30 + 90$

AF19-024
Question:
What is the slope of a line that goes through (7, 12) and (13, 84)?
A) 10
B) 11
C) 12
D) 72

AF19-025
Question:

Simplify: $68x + 19 - 57x - 34$
A) $11x - 15$
B) $125x - 15$
C) $11x + 53$
D) $125x + 53$

AF19-026
Question:
If $g(x) = x^2 - 11x$, what is $g(15)$?
A) 45
B) 60
C) 165
D) 225

AF19-027
Question:
A sequence follows the rule “multiply by 4, then subtract 13.” If the first term is 11, what is the second term?
A) 31
B) 35
C) 44
D) 57

AF19-028
Question:
Solve: $22(x - 4) + 18 = 172$
A) 9
B) 10
C) 11
D) 12

AF19-029
Question:
A line crosses the y-axis at -19 and rises 9 units for every 1 unit to the right. Which equation represents the line?
A) $y = -19x + 9$
B) $y = 9x - 19$
C) $y = -9x - 19$
D) $y = 19x + 9$

AF19-030
Question:
If $26/39 = x/117$, what is x?
A) 65
B) 72
C) 78
D) 84

AF19-031
Question:
Solve: $-30x + 120 = -240$

A) -12
B) -8
C) 8
D) 12

AF19-032
Question:
A warehouse starts with 4,200 labels. Each package uses 210 labels. Which expression gives the number of labels left after p packages?
A) 4,200 + 210p
B) 4,200 - 210p
C) 210p - 4,200
D) 4,200 ÷ 210p

AF19-033
Question:
Which equation has x = 25 as its solution?
A) 2x + 17 = 65
B) 4x - 22 = 78
C) x/5 + 8 = 12
D) 3x - 11 = 62

PART B — ANSWER KEY + EXPLANATIONS

AF19-001 — Correct: B — Explanation:
Solve 21x - 15 = 111. Add 15 to both sides: 21x = 126. Divide by 21: x = 6. The most common mistake is subtracting 15 instead of adding 15.

AF19-002 — Correct: C — Explanation:
Start with 19n + 8 = 141. Subtract 8 from both sides: 19n = 133. Divide by 19: n = 7. A common mistake is stopping at 133 and forgetting to divide by 19.

AF19-003 — Correct: A — Explanation:
Distribute first: 6(5x - 3) = 30x - 18. Then subtract 8x: 30x - 8x - 18 = 22x - 18. A common mistake is forgetting to multiply -3 by 6.

AF19-004 — Correct: A — Explanation:
The company charges $38 for each job, so that part is 38j. The $30 setup fee is added one time. The total charge is 38j + 30. A common mistake is treating the setup fee as the per-job charge.

AF19-005 — Correct: B — Explanation:
Substitute 8 for x: f(8) = 16(8) - 41 = 128 - 41 = 87. A common mistake is adding 41 instead of subtracting it.

AF19-006 — Correct: C — Explanation:
Test the choices in y = 14x + 2. For (3, 44), 14(3) + 2 = 42 + 2 = 44, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF19-007 — Correct: C — Explanation:
Solve 22(x - 5) = 154. Divide both sides by 22: x - 5 = 7. Add 5: x = 12. A common mistake is adding 5 before undoing the multiplication by 22.

AF19-008 — Correct: C — Explanation:
The pattern increases by 18 each time: 8, 26, 44, 62. Add 18 to 62: 80. A common mistake is assuming the increase is 17 because the numbers are close together.

AF19-009 — Correct: C — Explanation:
$x/35 = 3$ means x divided by 35 equals 3. Multiply both sides by 35: $x = 105$. A common mistake is adding 35 and 3 instead of multiplying.

AF19-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -16 and the y-intercept is 13, so the equation is $y = -16x + 13$. A common mistake is switching the slope and y-intercept.

AF19-011 — Correct: C — Explanation:
Find the unit rate: 264 parts ÷ 8 bins = 33 parts per bin. For 5 bins: $33 \times 5 = 165$. A common mistake is multiplying 264 by 5 without finding the rate first.

AF19-012 — Correct: B — Explanation:
Solve $240 - 26x = 136$. Subtract 240 from both sides: $-26x = -104$. Divide by -26: $x = 4$. A common mistake is losing the negative sign after subtracting 240.

AF19-013 — Correct: A — Explanation:
Combine like terms: $90x - 43x = 47x$. The constant 12 stays the same. The expression becomes $47x + 12$. A common mistake is combining the constant with the variable terms.

AF19-014 — Correct: C — Explanation:
The shelf starts with 1,300 parts. Each order removes 65 parts, so o orders remove 65o parts. The number left is $P = 1{,}300 - 65o$. A common mistake is adding 65o even though parts are being removed.

AF19-015 — Correct: A — Explanation:
Substitute $x = 11$ into $y = -20x + 260$: $y = -20(11) + 260 = -220 + 260 = 40$. A common mistake is treating -20(11) as positive 220.

AF19-016 — Correct: C — Explanation:
Solve $32x - 50 = 21x + 60$. Subtract 21x: $11x - 50 = 60$. Add 50: $11x = 110$. Divide by 11: $x = 10$. A common mistake is adding 50 to only one side.

AF19-017 — Correct: C — Explanation:
Solve $17x - 6 > 164$. Add 6: $17x > 170$. Divide by 17: $x > 10$. The only listed value greater than 10 is 11. A common mistake is choosing 10, but $17(10) - 6 = 164$, not greater than 164.

AF19-018 — Correct: A — Explanation:
The y-values increase by 19 each time x increases by 1, so the slope is 19. When $x = 0$, $y = -9$, so the y-intercept is -9. The rule is $y = 19x - 9$. A common mistake is using +9 instead of -9.

AF19-019 — Correct: C — Explanation:
Distribute: $16x + 80 = 13x + 116$. Subtract 13x: $3x + 80 = 116$. Subtract 80: $3x = 36$. Divide by 3: $x = 12$. A common mistake is forgetting to multiply 5 by 16.

AF19-020 — Correct: C — Explanation:
Substitute $a = -13$ and $b = 11$: $3a + 4b = 3(-13) + 4(11) = -39 + 44 = 5$. A common mistake is treating $3(-13)$ as positive 39.

AF19-021 — Correct: C — Explanation:
Quadrant III has a negative x-coordinate and a negative y-coordinate. The point (-15, -8) fits this description. A common mistake is choosing (-15, 8), which is Quadrant II.

AF19-022 — Correct: C — Explanation:
210 items in 14 minutes means $210 \div 14 = 15$ items per minute. In 9 minutes: $15 \times 9 = 135$. A common mistake is multiplying 210 by 9 without finding the rate first.

AF19-023 — Correct: B — Explanation:
Start with $B = 30r + 90$. Subtract 90 from both sides: $B - 90 = 30r$. Divide by 30: $r = (B - 90)/30$. A common mistake is dividing B by 30 first and then subtracting 90.

AF19-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (7, 12) to (13, 84), y increases by 72 and x increases by 6. The slope is $72 \div 6 = 12$. A common mistake is using 72 as the slope without dividing by the change in x.

AF19-025 — Correct: A — Explanation:
Combine like terms: $68x - 57x = 11x$, and $19 - 34 = -15$. The simplified expression is $11x - 15$. A common mistake is writing $11x + 15$ because the negative sign before 34 is missed.

AF19-026 — Correct: B — Explanation:
Substitute 15 for x: $g(15) = 15^2 - 11(15) = 225 - 165 = 60$. A common mistake is calculating 15^2 as 30 instead of 225.

AF19-027 — Correct: A — Explanation:
Start with 11. Multiply by 4: 44. Then subtract 13: 31. A common mistake is subtracting 13 first and then multiplying.

AF19-028 — Correct: C — Explanation:
Solve $22(x - 4) + 18 = 172$. Distribute: $22x - 88 + 18 = 172$. Combine: $22x - 70 = 172$. Add 70: $22x = 242$. Divide by 22: $x = 11$. A common mistake is forgetting to combine -88 and +18.

AF19-029 — Correct: B — Explanation:
The line rises 9 units for every 1 unit to the right, so the slope is 9. It crosses the y-axis at -19, so the equation is $y = 9x - 19$. A common mistake is switching the slope and intercept.

AF19-030 — Correct: C — Explanation:
Use the proportion $26/39 = x/117$. Since $39 \times 3 = 117$, multiply 26 by 3: $x = 78$. A common mistake is adding 78 to the numerator because 39 becomes 117.

AF19-031 — Correct: D — Explanation:
Solve $-30x + 120 = -240$. Subtract 120 from both sides: $-30x = -360$. Divide by -30: $x = 12$. A common mistake is giving -12 because both sides contain negative numbers.

AF19-032 — Correct: B — Explanation:
The warehouse starts with 4,200 labels. Each package uses 210 labels, so p packages use 210p

labels. The number left is 4,200 - 210p. A common mistake is adding 210p even though labels are being used.

AF19-033 — Correct: B — Explanation:
Test $x = 25$. In choice B, $4(25) - 22 = 100 - 22 = 78$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.